Breaking the tunnel magnetoresistance plateau: Achieving high performance in magnetic tunnel junctions using advanced interface and barrier materials design

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I present our recent advances in magnetic tunnel junctions (MTJs), focusing on breaking the tunnel magnetoresistance (TMR) ratio record and eliminating material design limitation through advanced barrier interface and material developments. We achieved a record TMR ratio of 631% at room temperature using a CoFe/MgO/CoFe(001) epitaxial MTJ structure with precisely engineered atomic-level MgO barrier interfaces. We are further exploring low-bandgap barriers to reduce resistance area products of MTJs. We developed a high-entropy oxide (HEO) barrier, LiTiMgAlGaO, that resulted in a low barrier height and a relatively high TMR ratio. Therefore, HEO barriers are a promising platform for highly scalable and reliable MTJ applications.

***Index Terms*—Spintronics, Tunneling magnetoresistance, magnetic tunnel junction, epitaxial growth, high entropy oxides**

I. Introduction

Magnetic tunnel junctions (MTJs) are key elements in spintronic devices such as hard disk (HDD) readers, magnetoresistive random access memory (MRAM) cells, and high-sensitivity magnetic sensors. One of the most important MTJ performance metrics is the tunnel magnetoresistance (TMR) ratio, which directly determines the electrical signal output of MTJs. The discovery of crystalline MgO barriers and the resulting coherent tunneling effect has dramatically increased the TMR ratio at room temperature, establishing CoFeB/MgO/CoFeB as the de facto standard for practical devices [1]. However, further enhancement of the TMR ratio in current device structures is now approaching its limit. Applications require ultrathin MgO barriers to minimize the resistance-area product (*RA*) for HDD readers [2] and strong perpendicular magnetic anisotropy in ferromagnetic layers spin-transfer-torque (STT) MRAMs [3]. Furthermore, the stringent demands regarding thermal stability and reliability for MRAM and sensor applications necessitate a significant increase in room temperature TMR ratios. Enhancing the TMR ratio through innovative interface and materials engineering has become an urgent issue in the spintronic devices. This presentation will discuss how advanced interface and barrier engineering led to the first improvement in room-temperature TMR ratios in 15 years, as well as the development of a new, low-resistance barrier material: the high-entropy oxide (HEO) barrier.

II. TMR Record Progress and Giant TMR oscillation

It has become well established that the TMR ratio is significantly influenced by the interface states at the ferromagnetic layers and the barrier. For this reason, we have refocused on a simple epitaxial Fe/MgO/Fe(001)-based MTJ structure. Using an MgO substrate and an epitaxial Cr underlayer, we achieved an Fe(001) layer with extremely high (001) orientation and atomic flatness via sputtering. Furthermore, we improved the crystallinity of the MgO film by employing electron beam evaporation. Additionally, we combined various interface engineering techniques, such as introducing CoFe and Mg layers at the interfaces, as well as post-annealing and post-oxidation of the MgO layer. As a result, a room-temperature TMR of 631%, the highest value to date, was achieved in the CoFe/MgO/CoFe(001) MTJ structure [4]. These results demonstrate the continuing potential of interface and crystalline engineering. Figure 1 shows the trend of the room-temperature TMR ratio.

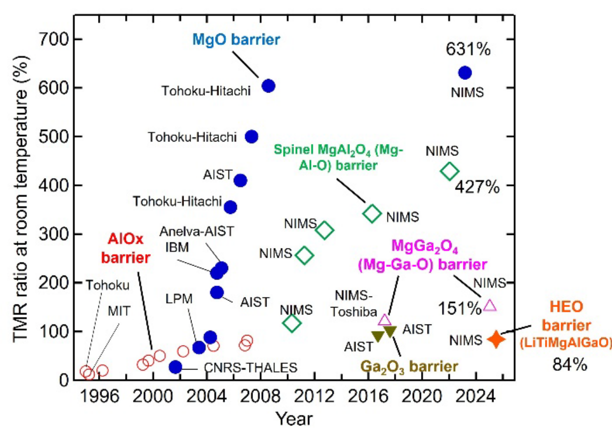

Fig. 1. History of room-temperature TMR ratio for different barrier materials.

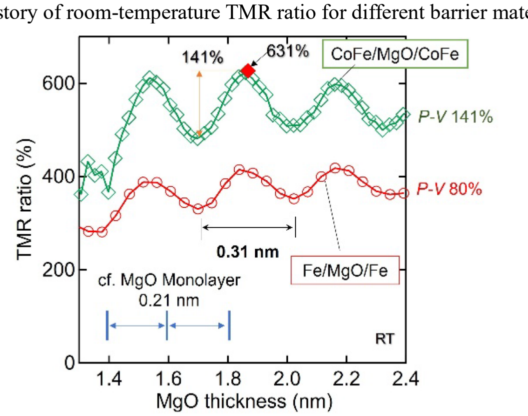

Fig. 2. Enhanced TMR oscillation effect observed in interface-engineered Fe/MgO/Fe [5] and CoFe/MgO/CoFe MTJs [4].

in tuning barrier material design for next-generation, high-performance, and ultra-high-density MTJ applications.

The improved TMR ratio enhances the TMR oscillation effect, which has a period of 0.3 nm as a function of the barrier thickness, as shown in Fig. 2 [4,5]. Because this oscillation has a period different from that of the MgO atomic layer thickness (0.21 nm), this mechanism had not been fully elucidated for many years. However, a recent theoretical study successfully reproduced the oscillation behavior by considering the superposition of the $\Delta_1\uparrow$ and $\Delta_2\downarrow$ states of the tunneling electrons [6].

III. Low-Resistance Barriers

A. Demand for low-resistance barriers

To achieve scaled MTJs for high-density devices, the barrier thickness must be minimized to reduce the *RA*. However, conventional MgO barriers face substantial TMR and crystallinity degradation when thinned below 1 nm. To address this issue, we propose using a semiconductor oxide barrier with a lower barrier height instead of the MgO barrier (Fig. 3). With this low resistance barrier, it becomes possible to increase the barrier thickness while maintaining the same *RA*. Consequently, improvements in the TMR ratio and reliability (breakdown voltage) in the low-*RA* region are expected.

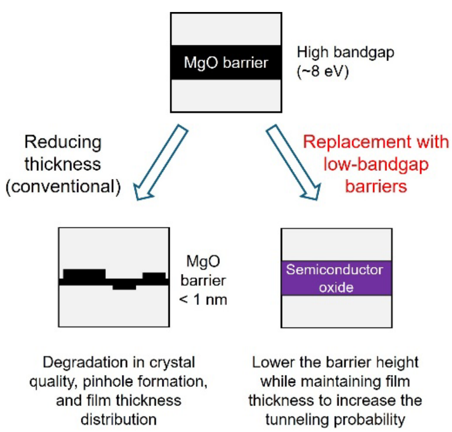

Fig. 3. Low resistance barrier strategy for future MTJs.

B. High-Entropy Barriers [7]

High-entropy oxide (HEO) barriers with multiple cations have been introduced as a new candidate for the low-resistance barriers. The presence of multiple cations has been reported to reduce the bandgap and enhance structural stability [8], raising expectations for their use as low-resistance barriers for MTJs. We recently developed an HEO barrier with a LiTiMgAlGaO composition using sputtering. The LiTiMgAlGaO film exhibits uniform cation distribution, as revealed by scanning transmission electron microscopy (STEM) and energy dispersive spectroscopy (EDS) analysis [Figs. 3(a) and (b)]. Notably, the LiTiMgAlGaO-based MTJ achieved a relatively high TMR ratios exceeding 80% at room temperature [Fig. 3(c)], indicating the presence of the coherent tunneling mechanism even with an HEO barrier. We also found that significant bias voltage dependence of resistance, revealing the low barrier height of LiTiMgAlGaO. This advance enables new flexibility

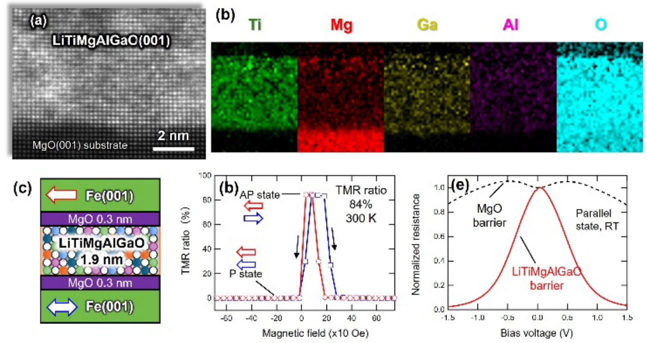

Fig. 4. High entropy oxide LiTiMgAlGaO thin film for MTJ barriers. (a) Cross-sectional STEM image of the LiTiMgAlGaO thin film. (b) Corresponding EDS elemental maps. (c) LiTiMgAlGaO-based MTJ structure. (d) TMR curves. (e) Bias voltage dependence of the resistance for MgO [5] and LiTiMgAlGaO-based MTJs. Some of the figures were taken from R.R Sihombing *et al.* [7], licensed under CC BY 4.0.

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